DAE-GSP: Discriminative Autoencoder With Gaussian Selective Patch for Multimodal Remote Sensing Image Classification

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***Abstract*— In the field of multimodal remote sensing image (MRSI) classification, self-supervised learning (SSL) algorithms have demonstrated significant advantages, particularly in scenarios with limited labeled samples. Existing SSL methods typically use auxiliary tasks within either contrastive or generative frameworks, focusing on discriminative or structural information separately. In this article, we propose a novel hybrid SSL paradigm, discriminative autoencoder with Gaussian selective patch (DAE-GSP) for MRSI classification. The DAE framework integrates contrastive learning with the masked image modeling (MIM) technique, allowing for simultaneous learning of structural information and discriminative representations from images. Furthermore, a cross-attention-based data-level fusion strategy is introduced during pretraining stage to enhance intermodal interactions, thereby improving the effectiveness of modality fusion. In addition, we propose a novel Gaussian selective patch (GSP) strategy, addressing the limitations of traditional square patch selection methods. Combined with self-supervised auxiliary tasks, this strategy facilitates the improved integration of multiple modalities and encourages the model to capture essential semantic information. Extensive experiments conducted on three public datasets (Houston2013, Augsburg, and Berlin) demonstrate the effectiveness of the proposed approach. With only ten labeled training samples per class, the proposed method achieves overall accuracy (OA) of 90.15%, 82.64%, and 71.03% on the Houston2013, Augsburg, and Berlin datasets, respectively, indicating improvements of 1.31%, 1.22%, and 1.48% over state-of-the-art methods.**

***Index Terms*— Contrastive learning, cross-attention, few-shot classification, generative self-supervised learning (SSL), multimodal images, patch selection strategy.**

I. Introduction

THE advent of remote sensing imaging technology has significantly broadened our understanding of the Earth's surface [1], offering unprecedented opportunities for monitoring, analysis, and prediction across various domains including agriculture [2], urban planning [3], environmental monitoring [4], and disaster management [5]. Recently, proliferation of multimodal remote sensing image (MRSI) data, encompassing optical, radar, thermal, and other modalities, brings new opportunities and challenges to this field. These data from different modalities offer unique insights due to their distinct properties and the nature of their interaction with terrestrial features [6], [7]. Multimodal learning, aiming to leverage the complementary strengths of these data to enhance the understanding and classification of remote sensing images, is therefore essential. The fusion of these varied data through multimodal learning not only enriches the feature space available for analysis but also mitigates the limitations of relying on a single data source. For example, the challenge of distinguishing ground objects with similar spectral responses in hyperspectral images (HSIs) can be effectively addressed by incorporating elevation information provided by light detection and ranging (LiDAR) data [8], [9], [10]. Meanwhile, the fusion of HSI and synthetic aperture radar (SAR) data is an effective strategy to mitigate the effects of inclement weather on HSI data and to reduce serious speckle noise on SAR data [11], [12], [13].

However, effective interpretation and understanding of the multimodal data remains a formidable challenge due to the unique properties of each modality. Meanwhile, acquiring labeled remote sensing data can be cost-prohibitive or infeasible due to accessibility issues or rapid environmental changes, and MRSI data from different modalities make annotation even more difficult and time-consuming. *How to efficiently process MRSI data in scenarios characterized by limited labeled data* becomes a key issue facing the field and has great research value.

Self-supervised learning (SSL) has emerged as a powerful paradigm to leverage the abundance of unlabeled data, enabling models to learn rich feature representations from a large number of unlabeled samples without the need for extensive annotation. By exploiting the inherent structure and correlation within multimodal remote sensing data, SSL can facilitate the extraction of meaningful representations that significantly enhance classification performance even with minimal labeled data. At this stage, mainstream SSL methods can be broadly categorized into two main types: contrastive and generative methods, each with its unique mechanisms, characteristic, and applications.

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Contrastive methods [14], [15], [16], [17] focus on learning by comparison and aim to learn representations that bring similar samples closer in the representation space while pushing dissimilar ones apart.

On the other hand, generative SSL is an emerging approach where models learn to generate or reconstruct input data as a way to understand its underlying structure without the need for explicit labels.

Following the idea of Bert [18] in natural language processing (NLP), MIM methods randomly mask a large part of image patches and use an autoencoder architecture to reconstruct the origin masked patch. This approach encourages the model to learn rich, contextual representations of the visual data, focusing on understanding the underlying structure and content of images.

Despite the excellent advancements of self-supervised-based MRSI classification methods mentioned above, there are still three issues.

1) *Auxiliary Tasks Bias:* As many previous researches suggest [19], [20], [21], [22], [23], contrastive learning largely relies on the argument methods (mainly referring to random cropping and flipping) and can capture image-level semantic features. So they will have excellent discriminative skills in separating different images while often lack spatial and spectral sensitivity to modeling the structure of each image [24], [25]. Meanwhile, MIM methods reconstruct the origin content of the masked data so that the model can learn semantic representation and structure information of an image [26] while often lack discriminate abilities compared with contrastive learning methods [23], [27], [28].
2) *Insufficient Fusion Strategies:* Currently, most MRSI classification methods use a Siamese network structure [29] where two modalities occupy different branch. Although the importance of fusion and interaction between modalities is emphasized [30], the fusion strategies adopted by SSL-based MRSI classification methods are relatively simple [31], [32]. How to effectively fuse multiple modalities in the pretraining stage is to be explored.
3) *Unfocused Square Patch Selection Strategy:* In previous works, as shown in Fig. 1(a), the input MRSI data are first split into overlapped square patches centered on the target pixel to balance computational complexity and prediction accuracy. However, objects in remote sensing images usually have extremely irregular shapes and the square patch may not contain effective contextual information or contain too much redundant information. In addition, overlapping patches may lead to information leakage or overfitting as a pixel would be included and processed multiple times in adjacent patches. More discussion can be found in Sections II-C and III-A.

The above problems will make the model learn less effective semantic information or contain bias during the pretraining stage, resulting in poor performance on downstream tasks. To efficiently process MRSI data in scenarios characterized by limited labeled data, we improved the self-supervised training strategy based on SS-MAE [32] and proposed discriminative autoencoder with Gaussian selective patch (DAE-GSP) for MRSI classification in the context of few-shot learning. Our work effectively combines contrastive learning and MIM methods to better extract structural information of images and discriminative information between images. To take full advantage of the hybrid self-supervised method, we further proposed a cross-attention-based data-level fusion strategy and a novel GSP strategy. The cross-attention-based module is mainly used to enhance the interaction between different branches of the Siamese network and encourages the fusion of data from different modalities. The GSP module can be regarded either as an effective data argument strategy for contrastive learning and MIM methods or as an enhancement component for receptive field and adaptability for irregular remote sensing objects without increasing computational complexity. The main contributions are summarized as follows.

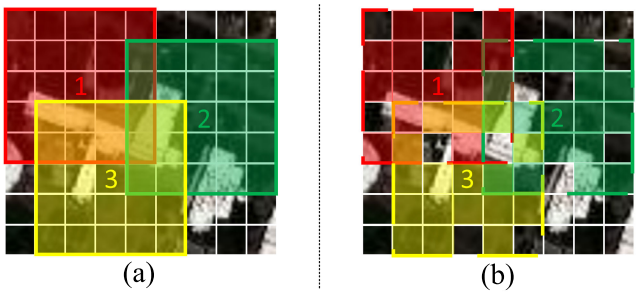


Fig. 1. Illustration of (a) traditional square patch selection and (b) proposed GSP.

1) A novel hybrid self-supervised framework, discriminative autoencoder (DAE), is proposed for MRSI classification. DAE effectively combines contrastive learning with MIM method to learn the structure representation of images and discriminative information between images with limited labeled data.
2) A cross-attention-based data-level fusion strategy is adopted in DAE to enhance the interaction between different branches and promote the fusion effect of multimodal data in the pretraining stage.
3) A novel Gaussian selective patch (GSP) strategy is proposed in cooperation with self-supervised auxiliary tasks to reduce computational complexity, increase receptive field, and alleviate overfitting.

The rest of this article is organized as follows. Related work regarding MRSI classification and two kinds of SSL is introduced in Section II. Section III gives the details of the proposed method. Extensive comparative experiments and ablation studies are conducted in Section IV, and the conclusion is presented in Section V.

II. Related Work

A. MRSI Classification

Many studies have explored various methodologies to effectively search for these rich treasures in the realm of MRSI classification. Previous methods can be roughly divided into three types according to the fusion strategy: data fusion, feature fusion, and decision fusion.

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Early MRSI classification methods were mainly based on decision-fusion. It involves making independent decisions from several classifiers and then combining their output to get the final result. Liao et al. [33] adopted four support vector machines (SVMs) to integrate morphological features from both principal components of HSI and LiDAR data to enhance multisensor data classification. Ge et al. [34] extracted extinction profiles and local binary patterns from both the data sources and then classified using collaborative representation-based classifier with Tikhonov regularization.

With the development of deep learning, feature fusion has become a more commonly used method. Chen et al. [35] introduced CNN to extract discriminant and invariant features from different modalities of remote sensing data. Recently, transformer and its variants have gained favor among researchers as an efficient feature extractor [36], [37], [38]. In addition to the model structure, many effective strategies were also used to improve the effect of feature fusion. Fang and Yan [39] used a knowledge graph to effectively use advanced semantic information and extract discriminative features to enhance MRSI classification. Multiscale [40], [41], [42] and multilevel [43], [44] features have also been shown to effectively enhance the feature extraction capabilities of the model. Moreover, text has also been used as a heterogeneous modality [45] or supervisory signal [46] to help capture rich inherent semantic information in complex feature of MRSI data.

Data fusion involves combining raw data from different modalities, which is often challenging but has good potential to offer a comprehensive view for subsequent analysis. At this stage, data-level fusion is still in the exploratory stage and is often used in conjunction with the first two methods. Li et al. [47] adopted a hierarchical structure to extract and use multiscale hierarchy features that encapsulate semantic information across different levels. Lin et al. [32] pad the input data of each modality and performed SSL in space and channels to learn representations of multimodal data.

B. SSL: Contrastive and Generative

SSL has garnered significant attention for its ability to leverage input data itself as supervision, benefiting a series of downstream tasks. SSL can be broadly categorized into contrastive and generative methods, each with its unique methodology, objectives, and nature.

Contrastive SSL aims to learn representations by distinguishing between similar (positive) and dissimilar (negative) pairs of data points. This approach often involves creating augmented versions of the same data point to serve as positive pairs and using different data points as negative pairs. CLIP [15] and SimCLR [14] demonstrate the power of using contrastive learning to learn representations. In the area of remote sensing, Feng et al. [31] use joint intra- and intermodal contrastive learning to encourage semantic consistency within and across modalities. Li et al. [48] focus on learning robust visual representations by gathering multilevel features and creating hard examples to enhance the discriminative ability of the model.

Generative SSL like MIM focuses on creating models that can generate or reconstruct inputs, enabling the model to learn representations by understanding the underlying structure of the data. He et al. [26] demonstrate significant advances in computer vision using a simple yet powerful paradigm of reconstructing images from partially masked inputs. In the field of remote sensing, due to the special properties of remote sensing images, the MIM method is still under research. Recently, Lin et al. [32] introduced a dual-branch approach that combines reconstructing masked spatial and spectral inputs and yielding considerable performance. Qu et al. [49] proposed a local diffusion shared-specific autoencoder (LDS^{2}AE), which incorporated masked training to the diffusion autoencoder to achieve local diffusion and captured the data distribution of different modalities.

C. Patch Selection Strategy

In traditional remote sensing image processing that involves HSI, an image is usually split into overlapped square patches in the shape of $p \times p \times c$ to balance computational complexity and prediction accuracy [17], [50]. This has also been carried over to the field of MRSI classification.

However, this operation has two shortcomings as shown in Fig. 1: on one hand, objects in remote sensing images usually have highly irregular shapes, making it difficult to determine the appropriate patch size to obtain useful contextual information. At the same time, square patches will easily lead to poor classification performance in classifying objects that rely on distant pixels, such as roads or rivers. On the other hand, since a pixel will be densely calculated multiple times by adjacent patches, the overlapped patches can easily result in overfitting and poor generalization performance.

III. Proposed Method

In this section, we will introduce the details of our proposed DAE-GSP. It includes a novel GSP strategy, two branches of the MIM model, and a cross-branch contrastive learning model. In our method, 3-D patches sampled from different modalities of RS images based on the GSP strategy are randomly masked from both spectral and spatial aspects. Visual patches are then embedded and sent to the encoder. The representation of masked embedding, which simultaneously contains discriminate information, is then projected based on the visual embedding through a cross-attention-based contrastive learning strategy. The decoder would use the representation to recover the masked part to complete the MIM method. The overview of the proposed method is shown in Fig. 2.

A. GSP Strategy

To solve the problems of traditional patch selection strategy, we propose selective patch strategy to cover more areas, reduce the amount of calculation, and alleviate overfitting. The detailed process is shown in Fig. 3. Given an MRSI dataset $D = \{H_i, M_i\}$, where $H_i \in R^{p_0 \times p_0 \times h_1}$ and $M_i \in R^{p_0 \times p_0 \times h_2}$. H_i and M_i are ith patches centered on the target pixel sampled from the same region but from different modalities of the

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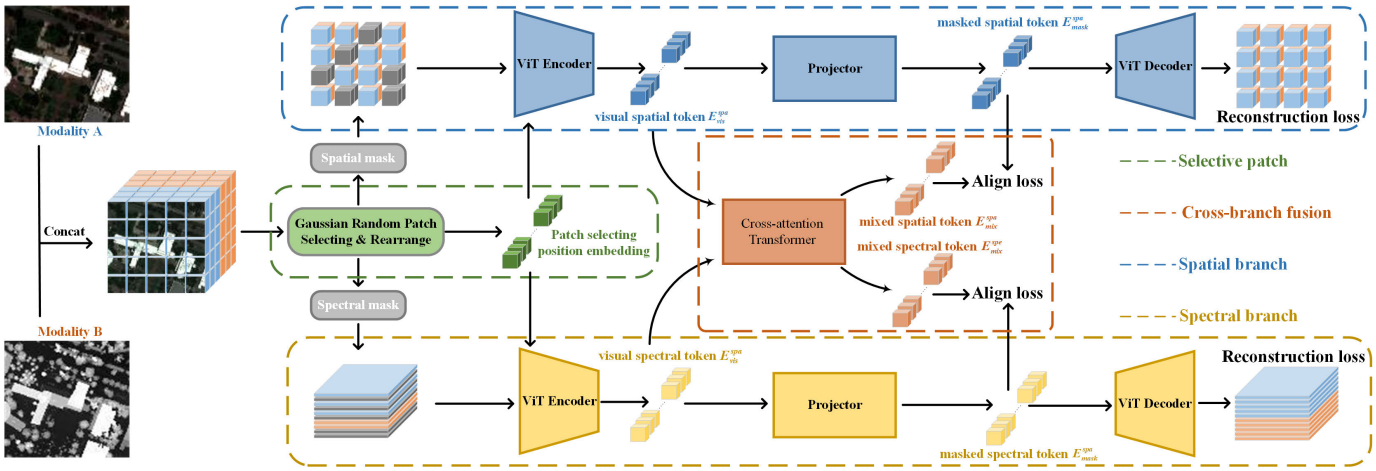


Fig. 2. Structure of DAE-GSP effectively amalgamates contrastive learning with the MIM and consists of four parts: GSP strategy, cross-branch contrastive fusion, and spatial and spectral reconstruction branch.

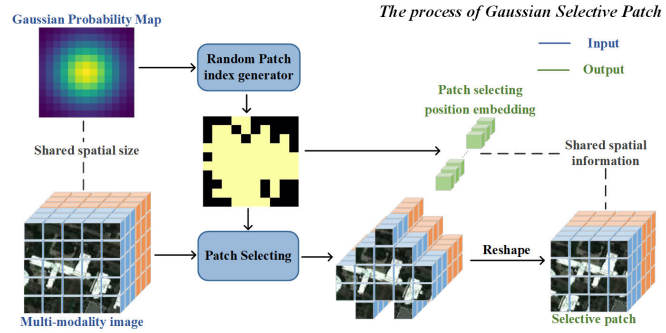


Fig. 3. Process of the proposed GSP strategy.

MRSI data. The patches are retrieved using a sliding window method and the patch size is p_0. It is worth noting that the patch here is different from the patch in the ViT encoder.

To get the selective patch, considering that remote sensing images are usually sensitive to distance, we use a modified 2-D Gaussian function as follows used as a probability map of a patch to select the pixels in it:

$$G_i(x, y) = \frac{1}{2\pi\sigma^2} e^{\frac{-(x-x_0+\text{temp})^2-(y-y_0+\text{temp})^2}{2\sigma^2}} \tag{1}$$

where x and y are the locations of a given pixel, x_0, and y_0 are the locations of the center pixel of a patch, temp is a hyperparameter that controls the probability of the pixels on the corners, and the value is set to 0.1. σ is the standard deviation which is a hyperparameter that controls the shape of the Gaussian probability map and the value is set to $p_0/4$.

We then normalize the probability map and select a certain amount of pixels based on it. The selected pixels will be reshaped into a new patch in the shape of $p_1 \times p_1 \times h_i$

$$\begin{aligned} M_i^{\text{sel}} &= \text{Reshape}(S(M_i, G_i)) \\ H_i^{\text{sel}} &= \text{Reshape}(S(H_i, G_i)) \end{aligned} \tag{2}$$

where S is the fixed-percentage weighted random sampling process based on the probability map and the selecting percentage is set to ${p_1}^2/{p_0}^2$. It is worth noting that we take out the key center pixel directly instead of processing it like other pixels.

To include the origin location information, we added an extra position embedding to each pixel. Cosine position embedding [51] is adopted to describe the relative positions' information of pixels

$$\text{PE}_{j,d} = \begin{cases} \cos\left(\dfrac{j}{10000^{2d/D}}\right), & d/2 = 0 \\ \sin\left(\dfrac{j}{10000^{2d/D}}\right), & d/2 \neq 0 \end{cases} \tag{3}$$

where j is the 1-D position of the pixel as [52] suggest there is no difference between using 1- or 2-D position embedding. d is the dth dimension of embedding and D is the overall dimension of embedding.

After adding the corresponding position embedding to the pixel of each modality, the image data from two modalities are concatenated along the channel dimension and used as the input of the model. The final input accepted by the model can be described as follows:

$$X_i = \text{pad}\left(M_i^{\text{sel}}, H_i^{\text{sel}}\right) + \text{PE}. \tag{4}$$

A graphical comparison of the proposed GSP and the traditional square patch selection strategy can be found in Fig. 1. Without too much calculation, GSP effectively assigns weights to different pixels based on distance and reduces the ratio of overlap pixels between patches. The former can effectively shift the attention of selected patches to the center. In this situation, with the same amount of computation, the patch can also be enlarged so that models would have the opportunity to acquire information of objects that rely on distant pixels, such as roads or rivers. The latter can reduce the times of processing the same pixel, thereby reducing the possibility of overfitting and ensuring more stable training.

B. Cross-Branch Contrastive Learning

Contrastive learning can make the model more discriminative for different categories. Some previous researches [31], [53], [54] have also explored the application of contrastive

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learning in MRSI classification. Inspired by [32], we introduced DAE with spatial and spectral reconstruction in each branch, respectively. On this basis, we proposed a cross-branch interaction based on cross-attention to make up for the shortcomings caused by insufficient feature fusion in previous studies and better complete the process of contrastive learning.

In the cross-branch contrastive learning model, a set of input X mentioned above are masked in space and spectrum, respectively, for each patch to get the visual spatial selective visual patch $X_{\text{vis}}^{\text{spa}}$ and spectral selective visual patch $X_{\text{vis}}^{\text{spe}}$ first. On the spatial branch, we adopt a masking strategy similar to masked autoencoder (MAE) while on the spectral branch a certain percentage of spectra are randomly masked. It is worth noting that since we need to mask a certain proportion of the spectrum, we assume that at least one modality has multiple channels. The mask strategy here is not only a part of MAE but also can be regarded as a data argument method for contrastive learning. As each pixel in remote sensing images represents a larger area and has fewer relationships with each other than traditional images, the masking percentage is only set to 30%.

Second, $X_{\text{vis}}^{\text{spa}}$ and $X_{\text{vis}}^{\text{spe}}$ are then fed into the corresponding branch to extract spatial/spectral information through a ViT encoder to get the visual spatial embedding $E_{\text{vis}}^{\text{spa}}$ and spectral embedding $E_{\text{vis}}^{\text{spe}}$. It is worth noting that after dimension conversion and a linear layer projecting, the encoder can share their weight to learn spatial and spectral features and thus reduce parameters without hurting performance. The process can be described as follows:

$$\begin{aligned} E_{\text{vis}}^{\text{spa}} &= g(\text{SpaMask}(X)) \\ E_{\text{vis}}^{\text{spe}} &= g(\text{SpeMask}(X)) \end{aligned} \tag{5}$$

where SpaMask and SpeMask represent randomly mask on the dimension of spatial and spectral, respectively. g represent the weight-shared spatial and spectral encoder.

Third, a projector is used to project visible spatial features $E_{\text{vis}}^{\text{spa}}$ and spectral features $E_{\text{vis}}^{\text{spe}}$ into masked spatial features $E_{\text{mask}}^{\text{spa}}$ and spectral features $E_{\text{mask}}^{\text{spe}}$, respectively. At the same time, we send the visual spatial features $E_{\text{vis}}^{\text{spa}}$ and spectral features $E_{\text{vis}}^{\text{spe}}$ into an encoder based on cross-attention to fuse the spatial and spectral information. Specifically, visual spatial features $E_{\text{vis}}^{\text{spa}}$ and spectral features $E_{\text{vis}}^{\text{spe}}$ are first mapped to query (Q), key (K), and value (V) using linear projection

$$\begin{aligned} Q^{\text{spa}} &= W_Q^{\text{spa}} E_{\text{vis}}^{\text{spa}}, \quad K^{\text{spa}} = W_K^{\text{spa}} E_{\text{vis}}^{\text{spa}}, \quad V^{\text{spa}} = W_V^{\text{spa}} E_{\text{vis}}^{\text{spa}} \\ Q^{\text{spe}} &= W_Q^{\text{spe}} E_{\text{vis}}^{\text{spe}}, \quad K^{\text{spe}} = W_K^{\text{spe}} E_{\text{vis}}^{\text{spe}}, \quad V^{\text{spe}} = W_V^{\text{spe}} E_{\text{vis}}^{\text{spe}}. \end{aligned} \tag{6}$$

Then, we can calculate the two-way attention weight

$$\begin{aligned} A^{\text{spa}\to\text{spe}} &= \text{Softmax}\left(\frac{Q^{\text{spa}}(K^{\text{spe}})^T}{\sqrt{d_k}}\right) \\ A^{\text{spe}\to\text{spa}} &= \text{Softmax}\left(\frac{Q^{\text{spe}}(K^{\text{spa}})^T}{\sqrt{d_k}}\right). \end{aligned} \tag{7}$$

Finally, we can calculate mixed visual spatial and spectral features with mutual information

$$\begin{aligned} E_{\text{mix}}^{\text{spa}} &= A^{\text{spa}\to\text{spe}} V^{\text{spa}} + E_{\text{vis}}^{\text{spa}} \\ E_{\text{mix}}^{\text{spe}} &= A^{\text{spe}\to\text{spa}} V^{\text{spe}} + E_{\text{vis}}^{\text{spe}}. \end{aligned} \tag{8}$$

After obtaining visual spatial and spectral features with mixed information, we can use contrastive learning to encourage the model to learn discriminative features and fuse spatial and spectral features for joint learning. We took an InfoNCE loss function [55] in a mini-batch as contrastive loss function formulated as follows:

$$\begin{aligned} L^{\text{spa}} &= -\log \frac{e^{s\left(E_{\text{mask}}^{\text{spa}}, E_{\text{mix}}^{\text{spa}+}\right)/\tau}}{e^{s\left(E_{\text{mask}}^{\text{spa}}, E_{\text{mix}}^{\text{spa}+}\right)/\tau} + \sum_{i=1}^{N-1} e^{s\left(E_{\text{mask}}^{\text{spa}}, E_{\text{mix}}^{\text{spa}-}\right)/\tau}} \\ L^{\text{spe}} &= -\log \frac{e^{s\left(E_{\text{mask}}^{\text{spe}}, E_{\text{mix}}^{\text{spe}+}\right)/\tau}}{e^{s\left(E_{\text{mask}}^{\text{spe}}, E_{\text{mix}}^{\text{spe}+}\right)/\tau} + \sum_{i=1}^{N-1} e^{s\left(E_{\text{mask}}^{\text{spe}}, E_{\text{mix}}^{\text{spe}-}\right)/\tau}} \end{aligned} \tag{9}$$

where $s(\cdot)$ is the cosine similarity function and N is the size of mini-batch. τ represents the temperature coefficient and its default value is 0.1. E_{mix}^{+} represents a positive sample, representing E_{mix} and E_{mask} belong to the same patch. E_{mix}^{-} represents a negative sample, representing E_{mix} is from other patches in a mini-batch.

The overall contrastive loss function of a mini-batch is as follows:

$$\mathcal{L}_{\text{align}} = \frac{1}{2N} \sum_{i=1}^{N} \left(L^{\text{spa}} + L^{\text{spe}}\right) \tag{10}$$

where N is the samples in a mini-batch during the pretraining stage.

C. Cross-Branch Reconstruction

After the predicted masked spatial features $E_{\text{mask}}^{\text{spa}}$ and spectral features $E_{\text{mask}}^{\text{spe}}$ are aligned via contrastive learning with the spatial–spectral hybrid features, they will have rich discriminative features and spatial–spectral information. $E_{\text{mask}}^{\text{spa}}$ and $E_{\text{mask}}^{\text{spe}}$ are then sent to the decoder to reconstruct the masked pixels or spectrum. Finally, we compare the pixels or spectral predicted by the model with the original masked pixels or spectral to obtain the reconstruction loss

$$\mathcal{L}_{\text{con}} = \mathcal{L}\left(T, R_{\text{spa}}\right) + \mathcal{L}\left(T, R_{\text{spe}}\right) \tag{11}$$

where T represents the original image, and R_{spa} and R_{spe} represent the reconstruct image of the masked part. Here, we use mean squared error (MSE) loss [56] for each loss function.

D. Overall Objective of Pretraining Stage

The final loss in the pretraining stage is a combination of contrastive loss and mask reconstruction loss. The overall loss function can be described as follows:

$$L = L_{\text{align}} + \beta L_{\text{con}} \tag{12}$$

where β is a hyperparameter to balance to influence of contrastive loss and mask reconstruction loss. We found that the value of β has almost no effect when the reconstruction loss and the contrast loss are about the same order of magnitude. We set β to 0.9 to adjust the values of reconstruction loss and contrastive loss to the same scale.

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TABLE I
NETWORK CONFIGURATION OF LOCAL FEATURE EXTRACTOR AND CLASSIFY HEADER

Modality	Local Feature Extractor			Classify Header	
	Block 1	Block 2	Block 3	Block 4	Block 5
HSI	1×1 Conv	$3 \times 3 \times 3$ Conv3D	1×1 Conv	FC	FC
	BN	BN	BN	-	-
	GELU	GELU	GELU	GELU	-
LiDAR/SAR	1×1 Conv	3×3 DWConv	1×1 Conv		
	BN	BN	BN		
	GELU	GELU	GELU		

E. Fine-Tune Stage

After the pretraining stage is completed, we can obtain an encoder (including the shared spectral–spatial encoder and cross-attention part) with discriminative capabilities and structural information extraction capabilities. The output of the encoder is concatenated along the feature axis and sent to the classification header to predict the category of the center pixel. In addition, as [32] suggests, we adopted two simple shallow CNN blocks and pad their output to the pretrained encoder's output to enrich the local pattern and improve the final classification accuracy. Each of the CNN blocks contains a projection layer, a convolution layer, and a projection layer. The details for the CNN blocks and classify header can be found in Table I. For HSI, the convolution layer is 3-D convolution layer and can be described as follows:

$$\begin{aligned} H_{\text{loc}} &= W_1(\mathcal{F}(W_2(H_i))) \\ \mathcal{F}(H_i) &= \text{Conv3D}(H_i) + H_i \end{aligned} \tag{13}$$

and for LiDAR/SAR, the convolution layer refers to a 3×3 depthwise convolution layer

$$\begin{aligned} M_{\text{loc}} &= W_1(\mathcal{F}(W_2(M_i))) \\ \mathcal{F}(M_i) &= \text{DWConv}(M_i) + M_i \end{aligned} \tag{14}$$

where W_1 and W_2 are a 1×1 convolution layer. We include the GELU activation and batch normalization after the convolution layers.

Finally, the features from the pretrained encoder and CNN blocks are stacked together and sent to a fully connected (FC) layer to get the results

$$t = \text{FC}\left(\text{pad}\left(E_{\text{vis}}^{\text{spa}}, E_{\text{vis}}^{\text{spe}}, E_{\text{mix}}^{\text{spa}}, E_{\text{mix}}^{\text{spe}}, H_{\text{loc}}, M_{\text{loc}}\right)\right). \tag{15}$$

We fine-tune the network on the small number of samples in the dataset, and the final classification loss is

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N_{\text{sup}}} \sum_{i=1}^{N_{\text{sup}}} y_i \log t_i \tag{16}$$

where N_{sup} are the training samples with labels in each mini-batch during the fine-tune stage. y_i is the predicted value of the ith training example and t_i is the corresponding ground truth.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Dataset

1) Houston2013 HSI-LiDAR Dataset: The dataset was from the 2013 IEEE GRSS data fusion contest.[1] It was captured using the ITRES CASI-1500 sensor, focusing on the University of Houston campus and surrounding areas. The HSI data encompass 144 spectral bands, with wavelengths ranging from 0.38 to 1.05 μm. The spatial dimensions of both the HSI and LiDAR data are 349×1905, boasting a spatial resolution of 2.5 m. Overall, the dataset categorizes the land into 15 distinct land cover classes.

2) Augsburg HSI-SAR Dataset [62]: This dataset originates from Augsburg, Germany, where data collection was performed using the HySpex sensor for HSI and the Sentinel-1 sensor for dual-polarimetric SAR (PolSAR). Both the datasets share spatial dimensions of 332×485, with a resolution of 30 m. The HSI data feature 180 spectral bands, covering wavelengths from 0.4 to 2.5 μm, while the PolSAR data comprise four distinct features. The dataset encompasses seven different land cover classes. The dataset can be accessed at website.[2]

3) Berlin HSI-SAR Dataset [62]: This dataset encompasses the geomorphological features of both urban and rural environments around Berlin. HSI data are derived from the synthesized EnMAP and HyMap HSI datasets, whereas SAR information comes from the Sentinel-1 dual-polarization (VV–VH) single-look complex (SLC) product provided by the European Space Agency (ESA). The HSI dataset consists of 797×220 pixels, with 244 spectral bands ranging from 0.4 to 2.5 μm and offers a spatial resolution of 30 m. Meanwhile, the SAR data present an image of 1723×476 pixels, with a spatial resolution of 13.89 m, and identifies eight distinct land cover classes. To match the SAR image size, the HSI data were processed through nearest neighbor interpolation. The dataset can be accessed at website.[2]

The visualized images of the three datasets are shown in Fig. 4, including the false-color image of HSI, gray-scale image of LiDAR, and SAR image.

B. Experimental Settings

1) Evaluation Metrics: In these experiments, the performance of MRSI classification is assessed by accuracy of each category, overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ). In addition to these commonly used metrics, to solve the problems of class imbalance and kappa coefficient that are difficult to interpret [63], [64], we introduced quantity disagreement (QD) [63], macro F1-score ($F1_{\text{macro}}$), and weighted F1-score ($F1_{\text{weight}}$) [65], [66] as additional metrics.

2) Implementation Details: All the experiments are implemented using the Python 3.8 and PyTorch framework on the computers with Ubuntu 22.04 LTS and NVIDIA RTX 3090 GPU with 24-GB memory. For our proposed model, in the hybrid self-supervised pretraining stage, 10%

[1] https://hyperspectral.ee.uh.edu/?page_id=459
[2] https://github.com/danfenghong/ISPRS_S2FL

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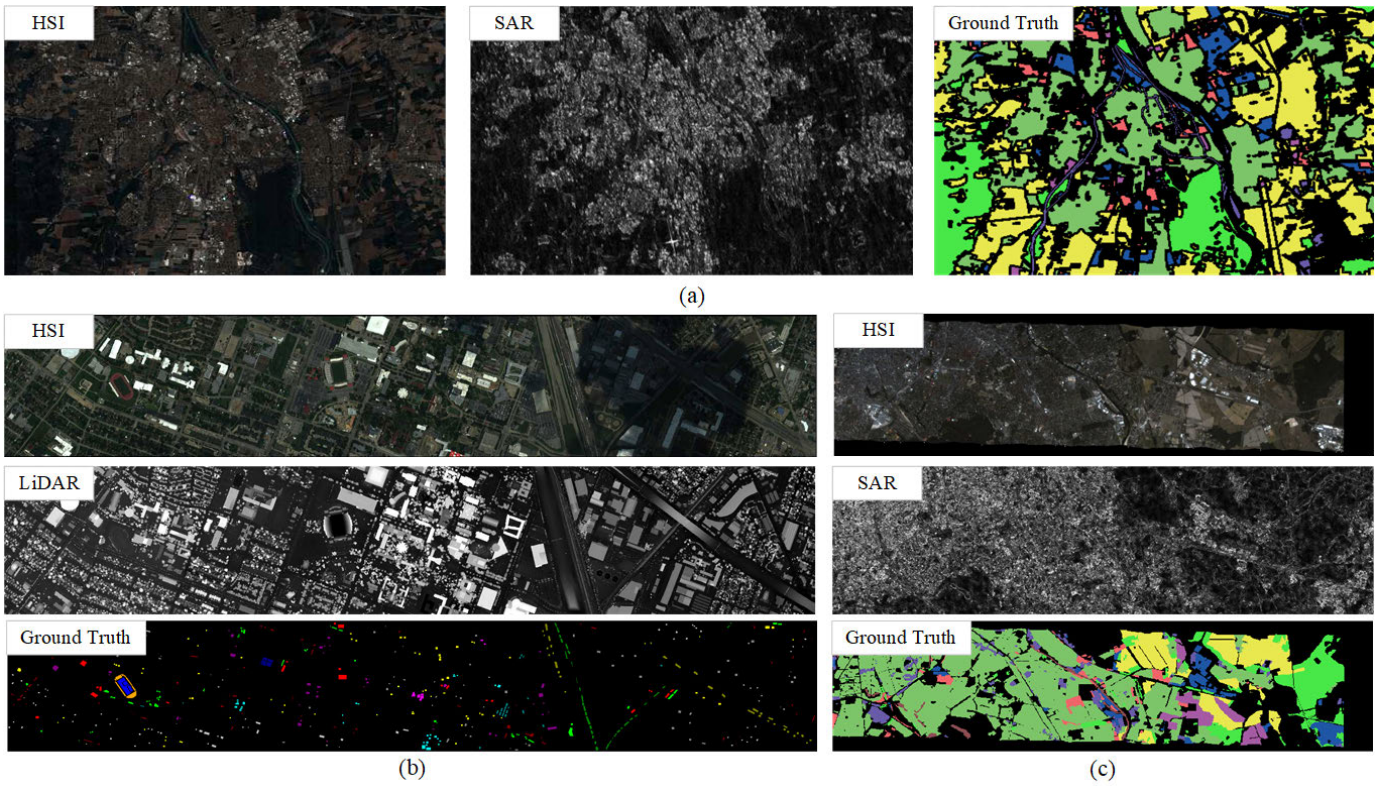


Fig. 4. Visualization of the MRSI datasets. (a) Augsburg dataset. (b) Houston 2013 dataset. (c) Berlin dataset.

of the pixels are selected as pretraining samples. In the fine-tune stage, ten labeled samples from each class are randomly selected as training samples from the origin training set and the rest is used as the testing set in the experiments. The details of the training samples and testing samples are listed in Table II. The batch size is set to 512, the learning rate is specified as 0.0001, and 300 epochs are executed for both the pretraining stage and fine-tune stage. Adamw [67] optimizer is applied to optimize the proposed model with a weight decay of 5e-2. Since our method is more of an improvement on the strategy, for the sake of simplicity and comparability, we directly use the same hyperparameter for the encoder and decoder in the previous study [32].

C. Comparison to State-of-the-Art Works

The proposed model's effectiveness is compared against a selection of representative state-of-the-art methods: NNC-Net [60], CMCL [31], SS-MAE [32], LDS2AE [49], and UACL [61]. In addition to these multimodal SSL methods, we also added two single-modal SSL methods and multi-modal supervised methods for comparison. The single-modal SSL methods include SimCLR [14] (with ResNet-50) and GraphGST [57]. The multimodal supervised learning methods include MIFNet [58] and NCGLF2 [59].

For single-modal SSL methods, SimCLR [14] is a classic contrastive learning framework that learns visual representations by maximizing the similarity between augmented versions of the same image and contrasting it with other images. GraphGST [57] integrates a transformer architecture with graph-based generative models and introduced cross-view contrastive learning.

For multimodal supervised learning methods, MIFNet [58] used a multiscale interactive fusion network to capture complementary information between the two modalities. NCGLF2 [59] fuses multisource remote sensing data by integrating global–local features and processing high–low-frequency information with a combination of CNN and transformer.

For multimodal SSL methods, NNCNet [60] applies nearest-neighbor-based contrastive learning to leverage the similarities between neighboring samples. CMCL [31] used joint intra- and cross-modal contrastive learning to better mine multimodal feature representations. SS-MAE [32] proposed a spatial–spectral masked autoencoder and exploited the spatial–spectral representations of the multimodal data. LDS2AE [49] introduced a local diffusion shared-specific autoencoder with masked training to capture the data distribution of different modalities and learn multimodal shared feature. UACL [61] focused on the problem of unreliable samples and proposed a novel uncertainty-aware contrastive learning method to learn more discriminative spectral–spatial features.

1) Quantitative Evaluation: The accuracy of each class, OA, AA, κ, QD, $F1_{\text{macro}}$, and $F1_{\text{weight}}$ of the three datasets for different methods is represented in Tables III–V. Note that these methods use the original parameters from this article. Each experiment is executed ten times with random seed from 0 to 9 so that the mean and standard deviation of each index are reported in the experiments.

Generally speaking, since self-supervised methods can learn effective semantic information from a large number of unlabeled samples, modern single-modal self-supervised methods (i.e., GraphGST) have better performance and stability

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TABLE II

LAND COVER CLASSES, NUMBER OF LABELED TRAINING AND TESTING SAMPLES, AND COLOR IN CLASSIFICATION MAP FOR THREE DATASETS, INCLUDING THE HOUSTON 2013, AUGSBURG, AND BERLIN DATASETS

Class No.	Houston2013				Augsburg				Berlin			
	Class name	Training	Testing	Color	Class name	Training	Testing	Color	Class name	Training	Testing	Color
1	Healthy Grass	10	1241		Forest	10	13497		Forest	10	54944	
2	Stressed Grass	10	1244		Residential Area	10	30319		Residential Area	10	268632	
3	Synthetic Grass	10	687		Industrial Area	10	3841		Industrial Area	10	19556	
4	Tree	10	1234		Low Plants	10	26847		Low Plants	10	59272	
5	Soil	10	1232		Allotment	10	565		Soil	10	17416	
6	Water	10	315		Commercial Area	10	1635		Allotment	10	13295	
7	Residential	10	1258		Water	10	1520		Commercial Area	10	24814	
8	Commercial	10	1234						Water	10	6662	
9	Road	10	1242									
10	Highway	10	1217									
11	Railway	10	1225									
12	Parking Lot1	10	1223									
13	Parking Lot2	10	459									
14	Tennis Court	10	418									
15	Running Track	10	650									
	Total	150	14879		Total	70	78224		Total	80	464591	

TABLE III

RESULTS OF DIFFERENT METHODS ON THE AUGSBURG DATASET, WITH THE BEST RESULT IN BOLD AND THE SECOND BEST RESULT UNDERLINED

Class	self-supervised&sigle-modal		supervised&multi-modal		self-supervised&multi-modal					
	SimCLR [14]	GraphGST [57]	MIFNet [58]	NCGLF2 [59]	NNCNet [60]	CMCL [31]	SS-MAE [32]	LDS2AE [49]	UACL [61]	proposed
1	83.74±6.71	79.76±7.53	82.48±9.40	**92.29**±3.47	81.66±14.67	84.65±10.29	89.23±5.24	90.16±4.49	88.44±5.46	90.46±4.67
2	78.10±5.11	92.44±2.22	89.15±7.79	94.65±3.75	81.61±3.75	93.44±4.03	93.85±1.63	**97.28**±1.87	91.17±3.15	92.25±2.97
3	27.85±5.42	59.08±7.58	52.29±10.40	**64.52**±14.55	57.74±6.24	62.47±5.03	59.72±5.85	47.22±9.45	54.49±5.76	63.66±8.36
4	82.21±8.86	90.44±5.07	91.81±7.00	96.88±1.92	**99.38**±0.62	99.24±1.44	93.21±3.36	95.08±2.29	91.35±3.70	93.73±3.33
5	6.12±3.20	8.66±3.62	4.79±1.71	9.38±3.88	**18.11**±5.85	6.49±1.64	12.16±3.67	10.39±4.35	13.26±5.47	12.00±2.56
6	5.38±2.16	19.19±4.47	14.53±6.04	17.22±6.31	20.53±1.98	**30.65**±7.05	26.47±5.68	20.95±5.34	19.91±3.52	27.13±8.57
7	4.82±1.88	9.36±2.79	22.83±12.73	34.49±21.23	21.52±7.42	40.42±7.22	32.09±11.83	33.47±23.15	38.98±19.41	**44.32**±11.28
OA ↑	61.96±2.13	78.71±3.05	66.08±6.61	77.37±5.19	79.41±3.26	79.16±1.90	81.42±2.73	78.33±2.50	80.88±2.65	**82.64**±1.78
AA ↑	43.56±1.86	56.93±3.48	63.25±2.04	68.88±4.64	53.29±3.11	59.62±0.84	**72.03**±1.05	70.69±2.67	64.33±3.06	72.01±1.64
κ×100 ↑	49.15±2.57	70.55±4.09	56.45±7.64	69.90±6.21	70.72±4.52	71.84±2.67	74.61±3.36	71.16±3.58	73.54±3.35	**76.04**±2.26
QD×100 ↓	18.72±2.36	8.84±1.81	23.69±5.88	16.47±5.72	13.68±3.42	15.10±1.90	9.82±3.67	14.62±2.96	**8.58**±3.06	8.74±2.51
$F1_{macro}$ ↑	39.13±1.38	51.29±2.18	49.07±2.53	55.05±3.47	50.78±2.77	59.67±1.16	59.76±2.60	56.06±1.84	57.09±2.79	**61.35**±3.21
$F1_{weight}$ ↑	66.41±1.99	80.85±3.11	71.85±6.01	81.54±4.12	79.61±3.40	82.29±3.02	83.86±1.87	82.29±1.73	82.76±1.82	**84.72**±1.31

than supervised methods (i.e., NCGLF2). Especially in the difficult case of only ten labeled samples per class, the results of supervised methods are not only relatively low but also have large variance, despite the integration of multimodal data. For example, the standard deviation of OA of NCGLF2 reached 5.19, indicating that the model is less stable. At the same time, combining the richness and complementarity of multimodal information, multimodal self-supervised methods can achieve better performance under such harsh conditions. Although the single-class accuracy has some uncertainty, most of the best and suboptimal methods belong to this category. For example, the OA of SS-MAE is 2.71% higher than GraghGST and 4.05% higher than NCGLF2 on the Augsburg dataset. Among these methods, the comprehensive scores of our proposed method achieved the best or second-best results on all three datasets.

Specifically, for the Augsburg dataset, as can be observed in Table III, the results vary widely between categories, and the results of categories 3 and 5–7 are relatively poor. We believe that this is the challenge caused by the lack of distinctive intraclass features and imbalanced class distribution in the dataset. From the table, we can see that our method improved the performance of these classes while maintaining good performance of the other classes. The improvement of feature representation for harder classes can be attributed to the model's ability to use discriminative and structural information. For comprehensive scores, our methods exhibit a slight decrease of 0.02% on AA compared with SS-MAE and an increase of 0.16 on QD compared with UACL. This suggests that our method still has some potential to be explored in accuracy for some specific class and class distribution. Despite this disadvantage, our proposed method exhibits considerable performance improvements. The OA improvement achieved by our method is 1.76%, 4.31%, 1.22%, 3.48%, and 3.23% compared with previous multimodal SSL methods. The proposed method also achieves the highest F1 scores, illustrating its ability to perform well across both major and minor classes. Moreover, the highest kappa coefficient (76.04) and relatively low QD (8.74) emphasize that the proposed method effectively minimizes both overestimation and underestimation errors across classes.

For the Houston2013 dataset, as shown in Table IV, the proposed method continues to deliver highly competitive performance and surpasses other methods in comprehensive

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TABLE IV

RESULTS OF DIFFERENT METHODS ON THE HOUSTON 2013 DATASET, WITH THE BEST RESULT IN BOLD AND THE SECOND BEST RESULT UNDERLINED

Class	self-supervised&sigle-modal		supervised&multi-modal		self-supervised&multi-modal					
	SimCLR [14]	GraphGST [57]	MIFNet [58]	NCGLF2 [59]	NNCNet [60]	CMCL [31]	SS-MAE [32]	LDS2AE [49]	UACL [61]	proposed
1	**94.45**±1.46	92.56±4.30	91.18±6.01	85.25±8.71	92.73±6.20	90.66±5.04	91.98±6.77	90.27±6.43	93.69±4.67	93.26±4.53
2	**93.97**±2.81	92.21±2.59	92.23±6.66	88.11±8.02	87.01±7.02	86.39±6.39	91.99±5.78	87.69±5.81	92.37±5.41	93.17±5.30
3	**99.54**±1.02	95.12±4.10	85.58±10.89	99.43±0.91	99.49±0.68	99.53±0.92	95.13±5.58	94.74±3.60	97.88±5.73	96.56±6.27
4	97.46±0.90	92.00±4.65	89.25±5.05	83.04±6.95	87.23±8.26	89.55±5.11	94.08±3.32	90.00±6.00	**98.44**±1.00	96.35±2.73
5	97.93±0.89	93.91±5.65	97.27±1.41	95.29±5.58	94.56±2.03	88.90±6.28	**98.63**±1.96	97.39±1.54	97.32±2.37	97.79±2.25
6	10.90±2.86	80.03±15.40	62.67±15.78	95.54±5.45	96.67±5.58	95.19±2.54	94.78±15.16	84.45±12.73	**99.38**±0.64	99.14±1.18
7	76.49±2.39	78.77±8.45	83.51±6.16	83.55±6.35	79.44±5.94	86.73±5.61	91.52±3.04	**92.68**±2.26	91.48±4.56	91.12±3.21
8	81.36±3.38	86.53±4.96	86.03±10.21	**90.18**±8.79	77.53±9.48	87.00±5.54	82.28±8.37	89.19±7.85	84.76±6.10	83.90±7.24
9	78.37±4.06	80.98±4.01	74.19±7.65	73.49±14.28	74.09±11.28	76.63±5.22	84.16±9.82	**86.46**±3.32	81.15±11.11	84.63±8.57
10	**89.84**±7.51	72.94±7.35	68.82±7.00	72.61±13.83	68.01±15.18	77.44±8.38	72.38±7.95	84.80±6.09	75.92±6.63	79.02±4.51
11	81.21±1.59	86.66±4.79	76.09±6.60	78.12±7.78	87.47±5.62	84.96±4.10	86.27±6.39	**90.46**±3.86	85.51±4.78	89.65±4.37
12	**86.60**±4.05	80.17±6.37	74.40±6.12	72.25±10.10	76.73±6.42	73.89±4.69	81.47±7.20	82.34±7.62	77.96±8.55	82.77±6.72
13	57.09±3.43	66.61±13.33	63.32±9.76	72.59±13.03	93.85±4.10	89.97±5.69	94.21±3.44	74.09±9.77	**94.91**±4.52	93.77±8.89
14	87.41±3.22	89.14±9.50	62.51±11.61	98.33±2.33	**98.68**±2.90	96.94±1.44	95.57±6.56	90.74±6.32	97.19±2.78	97.48±3.02
15	93.92±9.64	86.08±7.66	98.19±2.06	94.93±4.77	**98.87**±1.35	98.71±0.86	96.61±2.19	94.74±3.87	95.38±3.04	97.57±2.05
OA ↑	78.31±1.17	84.89±1.04	80.65±1.63	82.80±2.14	83.68±2.42	85.92±1.57	88.02±1.48	88.68±1.54	88.84±1.23	**90.15**±1.37
AA ↑	81.81±1.20	86.36±1.51	82.45±1.57	85.37±1.91	86.15±2.04	88.17±1.17	89.58±1.27	90.05±1.38	89.95±1.11	**91.33**±1.15
κ×100 ↑	76.67±1.23	83.61±1.13	79.13±1.75	81.41±2.31	82.28±2.63	84.77±1.70	87.00±1.60	87.78±1.66	87.93±1.34	**89.35**±1.47
QD×100 ↓	13.26±0.41	6.08±0.77	8.86±1.54	7.54±1.79	5.72±2.32	4.88±0.97	5.25±0.96	4.28±0.73	4.47±0.86	**3.85**±0.54
$F1_{macro}$ ↑	78.39±1.16	85.02±1.54	80.12±1.82	84.61±1.99	86.37±1.91	87.69±1.22	89.39±1.41	89.05±1.45	90.09±1.05	**91.29**±1.32
$F1_{weight}$ ↑	80.89±1.50	84.75±1.04	80.49±1.53	82.46±2.26	83.62±2.42	85.78±1.57	87.88±1.56	88.66±1.58	88.75±1.23	**90.06**±1.40

TABLE V

RESULTS OF DIFFERENT METHODS ON THE BERLIN DATASET, WITH THE BEST RESULT IN BOLD AND THE SECOND BEST RESULT UNDERLINED

Class	self-supervised&sigle-modal		supervised&multi-modal		self-supervised&multi-modal					
	SimCLR [14]	GraphGST [57]	MIFNet [58]	NCGLF2 [59]	NNCNet [60]	CMCL [31]	SS-MAE [32]	LDS2AE [49]	UACL [61]	proposed
1	60.52±10.95	81.61±5.02	79.31±8.54	71.58±8.82	60.95±8.19	77.28±5.82	**84.14**±6.44	76.20±9.08	81.93±9.88	83.77±6.00
2	84.39±1.98	**93.65**±0.91	91.64±1.83	92.05±1.39	90.12±1.70	87.55±1.34	91.32±2.52	89.02±2.93	90.48±2.07	93.05±2.11
3	45.51±11.48	27.97±4.26	29.59±8.67	**50.10**±10.95	27.28±5.22	44.04±9.87	36.73±9.35	28.55±11.77	39.89±7.62	40.63±5.94
4	62.03±4.10	69.15±7.04	82.18±6.18	81.58±10.62	63.17±2.97	**84.40**±4.00	64.10±5.67	79.28±4.94	79.65±9.54	70.94±2.14
5	43.80±5.33	41.80±5.32	60.02±7.65	61.96±14.39	68.99±7.22	**89.40**±6.48	62.02±4.62	59.23±17.10	55.86±16.73	60.92±5.05
6	13.36±2.62	**28.71**±11.38	21.06±8.55	19.92±5.29	17.21±3.65	12.82±1.95	16.79±2.56	14.19±13.71	19.50±6.14	21.19±2.24
7	22.12±2.76	23.08±6.50	21.05±4.83	**29.42**±6.00	24.07±2.60	21.71±3.42	21.61±7.13	21.93±10.20	25.89±8.16	21.20±6.88
8	7.47±1.45	**65.26**±18.05	19.01±6.16	16.81±7.36	19.38±5.45	50.14±15.50	62.27±6.55	19.25±10.03	48.71±24.26	60.74±6.13
OA ↑	55.65±3.22	69.05±1.68	61.72±4.83	65.62±3.64	64.95±3.32	66.09±2.47	68.79±3.73	69.55±2.56	69.12±3.18	**71.03**±2.64
AA ↑	47.34±2.19	64.45±2.47	65.17±2.16	66.49±2.39	55.47±2.77	58.42±2.30	61.32±3.86	56.38±4.93	65.00±3.45	**66.70**±2.27
κ×100 ↑	38.05±2.57	56.09±1.81	48.74±4.74	52.84±3.87	49.61±3.49	50.74±2.51	54.24±4.41	53.83±3.87	55.37±3.78	**58.05**±2.47
QD×100 ↓	21.23±4.86	16.88±2.19	24.67±6.12	21.54±4.37	15.29±4.34	17.66±3.44	14.56±5.22	15.59±3.76	15.92±4.17	**13.96**±3.22
$F1_{macro}$ ↑	40.74±1.76	55.94±1.93	51.77±3.03	54.00±2.52	47.63±2.65	48.96±5.70	54.90±3.14	47.00±4.03	55.68±4.83	**58.86**±1.10
$F1_{weight}$ ↑	60.16±2.47	71.81±1.35	66.40±3.69	69.28±2.94	67.59±2.57	69.77±1.49	71.59±2.80	70.65±2.29	71.69±3.00	**73.81**±1.90

scores. Although the results of a single class are not optimal, the results of each class are among the top few. There is no class preference that other methods may have. For example, although UACL achieved the best results in categories 4, 6, and 13 and led by a slight advantage (1%–2%) compared with the proposed method, it lagged behind by a large margin in categories 9–12 (3%–5%). This balance between classes is crucial for remote sensing tasks. In terms of comprehensive scores, our method achieves the best results in all metrics. This indicates that the proposed self-supervised framework not only excels in classifying a diverse range of land-cover types but also effectively performs well in terms of consistency and accurately captures class distribution.

On the Berlin dataset in Table V, which poses additional challenges due to its complex and imbalanced class distributions, the proposed method still achieves the best results and outperforms all compared methods. Similarly, although no single category is outstanding, they all maintain a good level, resulting in high comprehensive scores. The proposed method achieves the best OA of 71.03%, with an improvement of 1.48% over the state-of-art methods. The weighted F1 score of 73.81% and the macro F1 score of 58.86% reflect the model's capability to handle class imbalance while maintaining good precision and recall across all classes.

2) Visual Evaluation: The classification map of the three datasets generated by different methods is shown in Figs. 5–7. The semantic information provided by a large number of unlabeled samples allows self-supervised methods to produce more accurate and realistic classification maps. More importantly, the structure information provided by the generative method has been shown to be effective in distinguishing some similar classes. This is evident in the highlighted area of Fig. 5 when differentiating between industrial area (blue) and commercial area (red). Guided by both discriminative and structural information, our method achieves more actual visual results.

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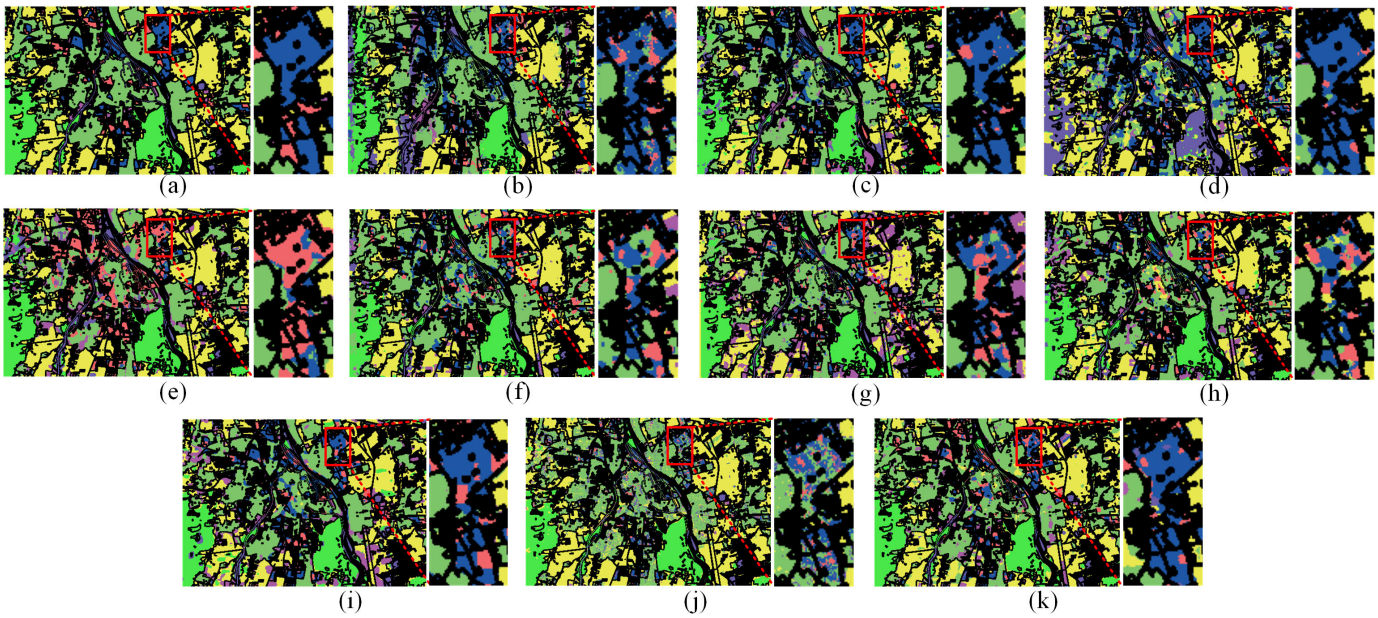


Fig. 5. Classification maps of different methods on the Augsburg dataset. (a) Ground truth. (b) SimCLR (OA = 61.96%). (c) GraphGST (OA = 78.71%). (d) MIFNet (OA = 66.08%). (e) NCGLF2 (OA = 77.37%). (f) NNCNet (OA = 79.41%). (g) CMCL (OA = 79.16%). (h) SS-MAE (OA = 81.42%). (i) LDS2AE (OA = 78.33%). (j) UACL (OA = 80.88%). (k) Proposed (OA = 82.64%).

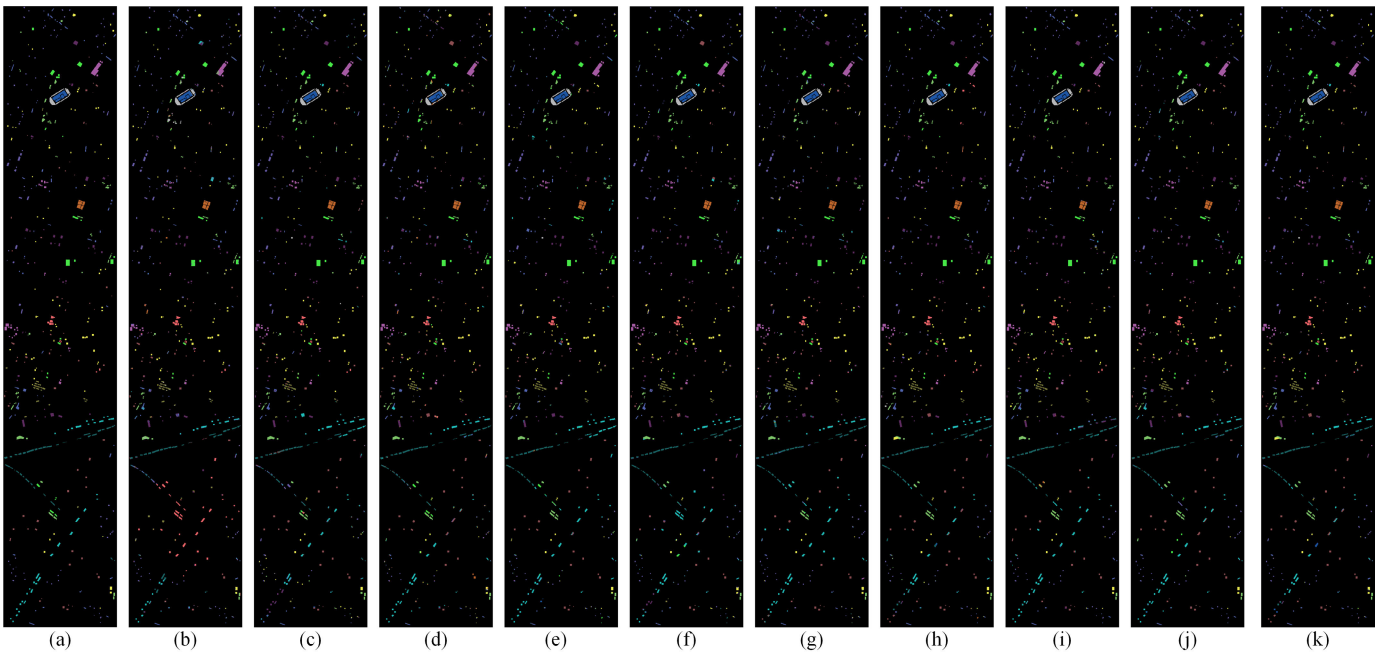


Fig. 6. Classification maps of different methods on the Houston 2013 dataset. (a) Ground truth. (b) SimCLR (OA = 78.31%). (c) GraphGST (OA = 84.89%). (d) MIFNet (OA = 80.65%). (e) NCGLF2 (OA = 82.80%). (f) NNCNet (OA = 83.68%). (g) CMCL (OA = 85.92%). (h) SS-MAE (OA = 88.02%). (i) LDS2AE (OA = 88.68%). (j) UACL (OA = 88.84%). (k) Proposed (OA = 90.15%).

D. Ablation Studies

To verify the effectiveness of our proposed method, we conducted a series of ablation experiments.

1) Effectiveness of Essential Components: In the pretraining stage, we adopted a hybrid self-supervised auxiliary task, a cross-attention-based data-level fusion strategy, and GSP. The validity of the proposed models was evaluated on the Houston2013 and Augsburg datasets, and the results can be found in Table VI. From the table, we can observe that the improvement brought by GSP is greater in Houston2013, while the improvement brought by DAE is greater in Augsburg due to the characteristics of their respective labeled data distribution. As shown in Fig. 4, the Housto2013 dataset has more sparsely labeled samples. As we used 10% randomly chosen samples during the self-supervised pretraining stage (e.g., DAE), the semantic information obtained from the model would be more similar to all samples than these

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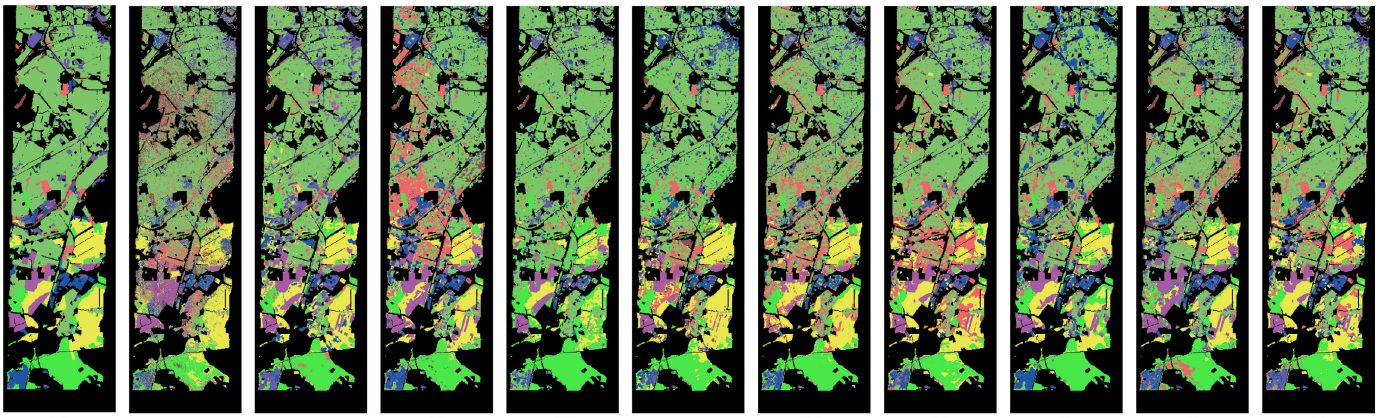


Fig. 7. Classification maps of different methods on the Berlin dataset. (a) Ground truth. (b) SimCLR (OA = 55.65%). (c) GraphGST (OA = 69.05%). (d) MIFNet (OA = 61.72%). (e) NCGLF2 (OA = 65.62%). (f) NNCNet (OA = 64.95%). (g) CMCL (OA = 66.09%). (h) SS-MAE (OA = 68.79%). (i) LDS2AE (OA = 69.55%). (j) UACL (OA = 69.12%). (k) Proposed (OA = 71.03%).

TABLE VI
EFFECTIVENESS OF ESSENTIAL COMPONENTS

Contrastive loss	Cross-branch fusion	Gaussian selective patch	Houston2013	Augsburg
			88.42	78.8
✓			88.43(+0.01)	79.86(+1.06)
✓	✓		89.11(+0.69)	80.00(+1.2)
		✓	89.05(+0.63)	79.21 (+0.41)
✓	✓	✓	89.52(+1.1)	80.87(+2.07)

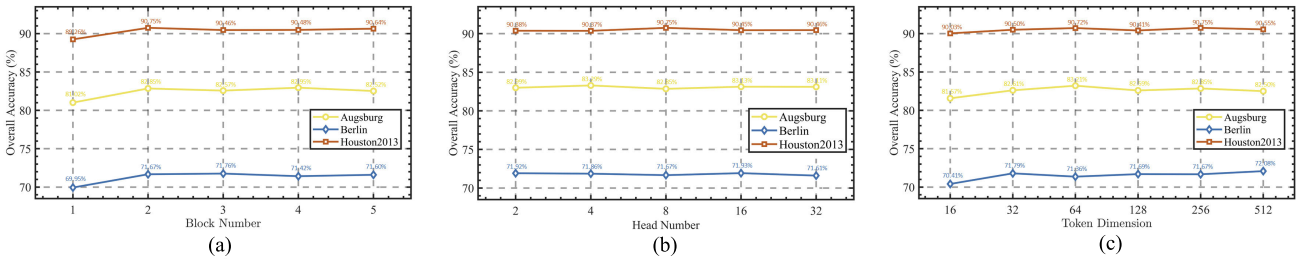


Fig. 8. Impact of model hyperparameters. Impact of (a) transformer block number, (b) transformer attention head number, and (c) transformer token dimension.

labeled samples. The distribution gap between all samples and human-labeled samples will lead to a smaller improvement for DAE on the Houston2013 dataset. Overall, each module improves the classification ability of the model across the datasets.

2) Impact of Model Hyperparameter: We studied the influence of model parameter settings on the performance of remote sensing image classification tasks (including block number, attention head number, and token dimension of the ViT encoder and decoder). To speed up the experiments, we set the hyperparameters of encoder and decoder to the same value. The results can be found in Fig. 8.

For the number of ViT blocks as shown in Fig. 8(a), the performance is relatively poor when its value is 1. We believe this is because the network layer is too shallow and the model cannot learn good features during pretraining and fine-tuning stages. As the number of block layers increases, the performance tends to be stable, with fluctuations within 1%. Our model achieved the highest OA values when the block number is 4 (82.95%), 3 (71.76%), and 2 (90.75%) for the Augsburg, Berlin, and Houston2013 datasets, respectively.

Fig. 8(b) shows the effect of different attention head numbers. It can be found that the number of attention headers has little impact on performance, with fluctuations within 1%. The performance peaks when the number of attention heads is 4 (83.29%), 16 (71.93%), and 8 (90.75%) for the Augsburg, Berlin, and Houston2013 datasets, respectively.

Fig. 8(c) reveals the relationship between different token dimension and classification performance. As can be seen from the figure, when the token dimension is too small, the feature expression ability of the model is limited. As a result, the model performs relatively poorly on the three datasets. As the token dimension increases, the three datasets would have fluctuating growth. For the Augsburg dataset, the performance peaks at a token dimension of 64 (83.21%), and then begins to decline. The Houston 2013 and Berlin datasets reach their maximums at 256 (90.75%) and 512 (72.08%), respectively. We believe this is due to the spatial resolution

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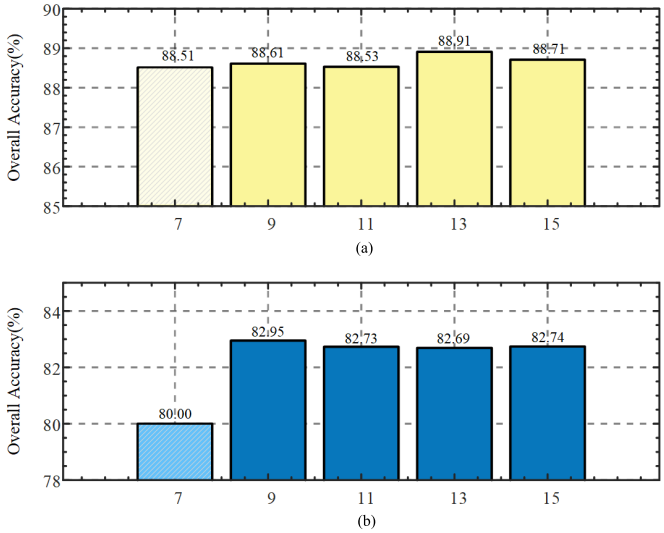


Fig. 9. Impact of different patch selections. (a) Houston2013. (b) Augsburg.

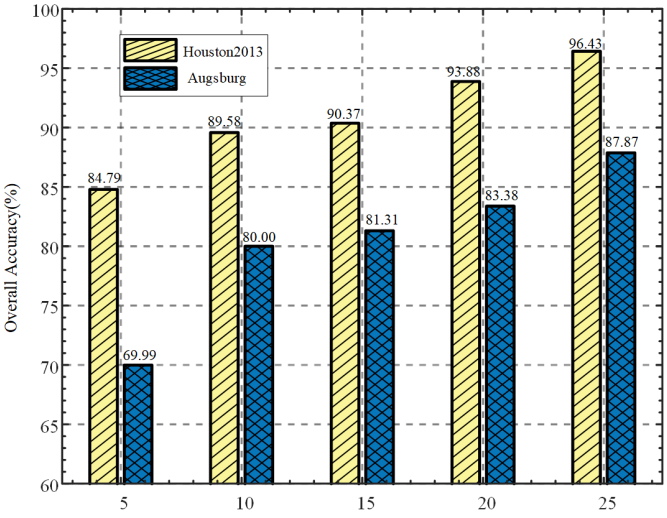


Fig. 10. Impact of the number of training samples in different datasets.

of the dataset itself. The 10% pretraining pixels on Augsburg (332 × 485) are less than that on Houston2013 (349 × 1905) and Berlin (1732 × 476). Therefore, using a smaller model on the Augsburg dataset can extract better semantic information, achieve better performance, and avoid overfitting.

3) Impact of Selective Patch: We conducted experiments with different patch selections on the Houston2013 and Augsburg datasets, and the results are shown in Fig. 9. We limit the final output patch size of the GSP module p_1 to 7 and control the GSP module's selection of patches by setting the GSP module's input size p_0. p_1 will affect the final computational complexity and p_0 will impact the receptive field of the model. We also add the result of conventional square patch selection with patch 7 for comparison to slightly different color. In general, the results of using GSP are better than that of using the traditional patch selection strategy. This proves that the proposed module can improve performance without increasing computational complexity. For different values of $p1$, the best-performing patch sizes vary slightly, with 13 being the best for Houston2013 and 9 for Augsburg. The performance gap between them is relatively small.

4) Impact of Different Numbers of Labeled Training Samples: To evaluate the classification performance of the proposed model varying the number of labeled training examples, we fine-tune our model by setting the number of labeled samples between 5 and 25 per class with intervals of 5. Fig. 10 illustrates the classification performance of different numbers of labeled training samples on the Augsburg and Houston2013 datasets. It is visible that our proposed model could achieve high classification accuracy even with a few labeled samples. This indicates that it can effectively extract features from multimodal data. The performance of the proposed model gradually increases with more training samples.

5) Analysis of Computational Complexity: We evaluated the complexity of the models using the number of model parameters and floating-point operations (FLOPs) on the Augsburg dataset. The comparison results are listed in Table VII. SimCLR uses ResNet-50 deep network as backbone so it has the largest number of parameters (27.803 MB) and relatively high FLOPs. For supervised multimodal methods (e.g., MIFNet and NCGLF2) and contrastive self-supervised methods (e.g., GraphGST, NNCNet, CMCL, and UACL), the models are relatively lightweight to strike a balance between handling complex multimodal data and avoiding overfitting. For generative methods such as SS-MAE, LDS2AE, and the proposed work, due to difficulty of auxiliary tasks, attention mechanism and transformer structure are usually adopted. As a result, the FLOPs is relatively high (despite the fact that LDS2AE used much fewer parameters). This is a future direction where we need to improve.

TABLE VII
EFFECTIVENESS OF ESSENTIAL COMPONENTS FOR THE AUGSBURG DATASET

model	Params. (MB)	FLOPs(MB)
SimCLR	27.803	37.152
GraphGST	1.765	1.769
MIFNet	0.173	11.713
NCGLF2	0.512	76.449
NNCNet	2.743	76.269
CMCL	0.281	8.413
SS-MAE	4.821	569.09
LDS2AE	0.722	140.178
UACL	0.188	18.605
DAE-GSP	7.60	696.53

V. LIMITATIONS AND FUTURE DIRECTIONS

Despite the good results, our method still has some shortcomings, which are also the directions we need to improve in the future. First, to extract useful semantic information from a large number of unlabeled samples, our method requires a long time for pretraining. How to conduct self-supervised pretraining tasks both efficiently and effectively is a topic of practical significance. In addition, auxiliary tasks in the existing methods (both contrastive and generative) mainly rely on the intrinsic relationship of pixel data. The design of auxiliary

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tasks usually lacks actual physical meaning (e.g. imaging) or semantic information. How to better design self-supervised auxiliary tasks is also an interesting topic.

VI. Conclusion

In this article, we proposed a novel hybrid SSL framework (DAE) and patch selection method (GSP). Both discriminative and structural information can be learned by the model from a large number of unlabeled samples to obtain better modal fusion effect and feature representation. The experimental results highlight the performance of our method in the scenarios if few-shot learning, with an improvement of 1.31%, 1.22%, and 1.48% on the Augsburg, Houston2013, and Berlin datasets, respectively. We hope that our work can bring some inspiration to related fields and practical applications facing the explosion and labeling difficulty of MRSI data.

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